

Rencana Proyek: Aplikasi Kalender & Penjadwalan (Laravel + Vue 3)

1) Ringkasan Eksekutif

Aplikasi kalender modern untuk pribadi & tim dengan dukungan penjadwalan pintar, event berulang (RRULE), notifikasi multi-kanal, sinkronisasi lintas perangkat, dan kolaborasi real-time. Target rilis **MVP** dalam 8 minggu dengan fokus pada UX ringan, performa cepat (<200ms cache hit, <800ms P95 API), serta keamanan tingkat produksi.

2) Tujuan Produk & KPI

- **KPI Adopsi:** 1.000 MAU, 200 DAU pada 30 hari pertama.
 - **KPI Kualitas:** Deliverability pengingat $\geq 99\%$, crash-free sessions $\geq 99.9\%$.
 - **KPI Performa:** P95 API < 800ms, P95 load kalender bulan < 1.5s.
 - **KPI Engagement:** Rata-rata 3+ event/user/minggu.
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3) Ruang Lingkup MVP

Fitur Inti 1. **Auth & Tenant:** Email+password, OAuth (Google/Microsoft, opsional), multi-tenant ringan (organisasi) dengan **role:** Owner, Admin, Member. 2. **Kalender:** Tampilan Hari/Minggu/Bulan/Agenda; drag-drop, resize event; kategori & warna; timezone-aware; multi-kalender per user. 3. **Event:** CRUD, peserta (internal/eksternal), lokasi, link meeting, lampiran, **recurrence** (RRULE + exceptions), deskripsi markdown. 4. **Penjadwalan:** Lihat **free/busy** peserta internal; rekomendasi slot waktu (rule-based v1) + buffer time. 5. **Notifikasi:** Pengingat via email, push web; jadwal berulang; quiet hours per user. 6. **Sinkronisasi:** Import/Export **ICS**; feed ICS read-only untuk share publik. 7. **Pencarian:** Event & peserta berbasis **Meilisearch** (full-text, typo-tolerant). 8. **Kolaborasi Real-time:** Update event & status RSVP via WebSocket.

Di Luar MVP (vNext): Integrasi Google/Microsoft two-way sync, reservasi resource (ruang/alat), CalDAV, mobile apps native, AI natural language ("meeting besok jam 3...").

4) Arsitektur Teknis

Frontend - **Vue 3 + TypeScript** (Vite), **Pinia** untuk state. - **UI:** Tailwind CSS + Headless UI / Radix-Vue alternatif; ikon Lucide. - **Kalender UI:** **FullCalendar** (Vue wrapper) dengan plugin timeGrid, interaction, rrule. - **i18n:** vue-i18n (default Bahasa Indonesia, dukung en-US). - **A11y:** Fokus, ARIA untuk dialog/modal, kontras.

Backend - **Laravel 11 (PHP 8.3)**, arsitektur modular (Domain-Driven ish), **Form Request** validation, **Resource/DTO** serializers. - **Auth: Laravel Sanctum** (SPA), Policy + Gate; **spatie/laravel-permission** untuk RBAC. - **Queue & Scheduler**: Laravel Queue (Redis) + **Horizon**; Laravel Scheduler + **spatie/laravel-schedule-monitor**. - **Realtime**: WebSocket via **Laravel Reverb** (opsional Pusher/Soketi). - **Search**: Laravel Scout + **Meilisearch**. - **Storage**: S3-compatible untuk lampiran. - **Observability**: Sentry (error), OpenTelemetry (tracing) → OTLP, Prometheus (metrics) + Grafana.

Data & Infrastruktur - **Database**: PostgreSQL 16. - **Cache**: Redis (rate limit, session, cache query). - **Infra**: Docker Compose (dev), prod via Docker + GH Actions, target deploy ke Fly.io/Render/VPS/K8s (opsional). - **Security**: Rate limiting, HTTPS, HSTS, CSP, rotasi secret, audit log.

5) Skema Data (ringkas)

users (id, name, email, password, timezone, locale)

organizations (id, name)

organization_user (org_id, user_id, role)

calendars (id, org_id, owner_user_id, name, color, is_default, is_public)

events (id, calendar_id, title, description_md, location, meeting_link, start_at, end_at, all_day, timezone, rrule, exdates, is_private)

event_participants (id, event_id, email, name, status[invited|accepted|declined|tentative], role[required|optional])

reminders (id, event_id, method[email|push], minutes_before)

free_busy_blocks (id, user_id, start_at, end_at, source[event|manual])

webhook_subscriptions (id, target_url, secret, events[event.created|updated|deleted])

audit_logs (id, actor_id, action, entity, entity_id, meta, created_at)

search_index → via Scout/Meili, tidak di DB.

Catatan: **rrule** menyimpan string RFC 5545 (contoh: `FREQ=WEEKLY;BYDAY=MO,WE;UNTIL=20251231`), **exdates** sebagai array tanggal pengecualian.

6) API Desain (REST)

Base: `/api/v1`

Auth - POST `/auth/login`, POST `/auth/logout`, GET `/auth/me`.

Kalender - GET `/calendars` (list), POST `/calendars`, GET `/calendars/{id}`, PATCH `/calendars/{id}`, DELETE `/calendars/{id}`.

Event - GET `/calendars/{id}/events?from=...&to=...` (range query, batched) - POST `/calendars/{id}/events` (create) - GET `/events/{id}` (instance master + expanded occurrences optional `?expand=true`) - PATCH `/events/{id}` (ubah satu seri/instance: `scope=series|single|future`) - DELETE `/events/{id}`

Penjadwalan - POST `/scheduling/free-busy` (input: peserta+rentang) → daftar slot sibuk - POST `/scheduling/suggest` (input: constraints) → rekomendasi slot

Notifikasi & RSVP - POST `/events/{id}/invite` - POST `/events/{id}/rsvp` (accept/decline/tentative)

Integrasi - GET `/calendars/{id}/ics` (feed publik) - POST `/import/ics` (upload ICS) - POST `/webhooks/subscribe`

Contoh Payload Event (ringkas)

```
{
  "title": "Weekly Standup",
  "start_at": "2025-09-05T09:00:00+07:00",
  "end_at": "2025-09-05T09:30:00+07:00",
  "timezone": "Asia/Jakarta",
  "rrule": "FREQ=WEEKLY;BYDAY=FR",
  "participants": [{ "email": "a@contoh.com", "role": "required" }],
  "reminders": [{ "method": "email", "minutes_before": 10 }]
```

7) UX Utama

- **Dashboard:** ringkasan event hari ini, tombol cepat "Buat Event".
- **Kalender:** Month/Week/Day/Agenda; drag-drop; shortcut; multi-select kalender; color coding.
- **Event Modal:** tab Detail, Peserta & RSVP, Pengingat, Berulang, Lampiran.
- **Scheduling Assistant:** tampilkan free/busy peserta; tombol "Suggest 3 slot".
- **Settings:** profil, timezone, notifikasi, integrasi, ICS feed, API token.

Prinsip: performa (lazy load/batch), offline-friendly (cache), aksesibilitas, tema gelap/terang.

8) Rencana Implementasi (8 Minggu)

Sprint 0 (Pra-work, 2-3 hari) - Repo, CI/CD (GitHub Actions), Docker dev, konvensi commit, lint/test.

Sprint 1 (Minggu 1-2) - Foundation - Auth (Sanctum), organisasi & RBAC. - Model + migrasi DB inti (calendars, events, participants, reminders). - FullCalendar integrasi dasar; tampilan bulanan.

Sprint 2 (Minggu 3-4) - Event & Recurrence - CRUD event lengkap, RRULE + exception, timezone handling. - Drag-drop & resize; warna kalender; search dasar (Scout+Meili). - Unit test domain (recurrence expansion, time math).

Sprint 3 (Minggu 5-6) - Notifikasi & Scheduling - Peningat (email/push) via Queue + Horizon. - Realtime (Reverb/Pusher) untuk update & RSVP. - Free/busy + Suggest slot (rule-based constraints: jam kerja, buffer). - ICS import/export + feed publik.

Sprint 4 (Minggu 7) - Kualitas & Hardening - Rate limiting, audit log, Sentry, OpenTelemetry. - A11y, i18n, tema gelap. - E2E test (Playwright) untuk alur utama.

Release (Minggu 8) - Beta - Load test, perf budget, observability dashboard. - Penyiapan staging & rilis produksi bertahap (feature flag).

Milestones: M1 Foundation, M2 Recurrence, M3 Realtime+Notif, M4 Beta.

9) Quality, Security, Compliance

- **Testing:** Unit (PHPUnit/Vitest), Feature (Laravel), E2E (Playwright). Target coverage 70%+ domain kritikal.
 - **Security:** OWASP Top 10, rate limit per IP/token, password hash Argon2id, 2FA (vNext), encrypt at rest (field sensitif), signed URL untuk ICS publik.
 - **Privacy:** Data minimization, retention policy, export delete data (GDPR-like), audit trail.
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10) Risiko & Mitigasi

- **Kompleksitas RRULE** → gunakan library rrule (frontend) & validasi server; uji kasus edge (DST, end-of-month).
 - **Timezone Drift** → simpan UTC + timezone; render di klien berdasarkan tz user.
 - **Deliverability Notif** → retry terjadwal (exponential backoff), fallback email jika push gagal.
 - **Skala** → batched fetch events per range; indeks pada (calendar_id, start_at); cache hasil expand.
 - **Integrasi Pihak Ketiga** → sandbox & feature flag; graceful degradation.
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11) Definisi Selesai (DoD)

- Lulus unit/feature/E2E, dokumentasi API (OpenAPI), tanpa error Sentry P1 di staging 72 jam, SLIs terpenuhi, a11y dasar terpenuhi.
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12) Backlog (estimasi SP)

- Auth + Org + RBAC (8)
 - CRUD Kalender (5)
 - CRUD Event + RRULE (13)
 - FullCalendar Integrasi (8)
 - Search Meili (5)
 - Notifikasi Email/Push (8)
 - Realtime RSVP/Update (8)
 - Free/Busy + Suggest Slot (8)
 - ICS Import/Export/Feed (8)
 - Audit Log + Rate Limit (5)
 - Observability (5)
 - Settings + ICS Token (5)
 - Theming + i18n (5)
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13) Contoh Teknis (cuplikan)

Migration Event (ringkas):

```
Schema::create('events', function (Blueprint $table) {
    $table->uuid('id')->primary();
    $table->uuid('calendar_id');
    $table->string('title');
    $table->text('description_md')->nullable();
    $table->string('location')->nullable();
    $table->string('meeting_link')->nullable();
    $table->timestampTz('start_at');
    $table->timestampTz('end_at');
    $table->boolean('all_day')->default(false);
    $table->string('timezone', 64)->default('UTC');
    $table->string('rrule')->nullable();
    $table->json('exdates')->nullable();
    $table->boolean('is_private')->default(false);
    $table->timestamps();
    $table->foreign('calendar_id')->references('id')->on('calendars');
```

```
$table->index(['calendar_id', 'start_at']);
});
```

Vue: Integrasi FullCalendar (sketsa):

```
<template>
  <FullCalendar
    :events="fetchEvents"
    :plugins="[dayGridPlugin, timeGridPlugin, interactionPlugin, rrulePlugin]"
    initialView="timeGridWeek"
    @dateClick="onDateClick"
    @eventDrop="onEventDrop"
  />
</template>
```

14) Tooling & DevEx

- **Lint/Format:** PHP-CS-Fixer, ESLint, Prettier, Stylelint.
- **Docs:** OpenAPI (Swagger UI), Storybook untuk komponen.
- **CI:** build, lint, test, DAST/ZAP scan, image SBOM.

15) Roadmap vNext

- Integrasi 2-arah Google/Microsoft + Webhook.
- Resource booking (ruang/alat) + approval flow.
- CalDAV server (opsional) untuk interoperabilitas.
- AI: natural language event creation & smart suggestions.
- Mobile PWA enhancements (offline create, push background sync).

16) Langkah Setup Dev (ringkas)

1. `cp .env.example .env` → set DB, Redis, mail, S3.
2. `composer install` & `php artisan key:generate`.
3. `php artisan migrate --seed`.
4. Jalankan Meilisearch & Redis (Docker), `php artisan horizon`.
5. `npm i` lalu `npm run dev`.

Catatan PM: fokus ke pengalaman penjadwalan cepat, minim klik, dan real-time feedback. Setiap fitur harus lolos uji edge-case timezone & recurrence sebelum rilis.